

CS1812

DISCRETE MATHEMATICS & SET THEORY



Module 1 : Sets

- Sets
- Set builder notations
- Types of Sets

Sets

A **set** is defined as a collection of distinct, well-defined objects. Each object in the set is known as an element of the set. We use curly brackets while writing a set. Sets are usually denoted by capital letters.

Ex :

$$A = \{2, 4, 6, 8\}$$

$$B = \{1, 3, 5, 7, 9, \dots\}$$

$$C = \{\text{January, February, March, } \dots\}$$

$$D = \{a, e, i, o, u\}$$

$$E = \text{Set of months having 32 days} = \{\}$$

Standard sets

Set of Natural Numbers (N) = $\{1, 2, 3, 4, \dots\}$

Set of Integers (I or Z) = $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

Set of Rational numbers (Q) = $\left\{\frac{p}{q} : \text{where } p \text{ and } q \text{ are integers and } q \neq 0\right\}$

Set of Irrational numbers (Q') = Numbers which are not rational

Set of Real numbers (R) = $Q \cup Q'$

Set of Complex numbers (C) = $\{a + ib : a \text{ and } b \text{ are real numbers and } i = \sqrt{-1}\}$

Elements of a Set

Elements or members are the terms or items or objects present in a set. They are enclosed in curly brackets and separated by commas. A set can have finite or infinite number of elements. To represent that an element is contained in a set, we use the symbol “ $\in$ ”. It is read as “belongs to”. If an element is not contained in a set, then we use the symbol “ $\notin$ ”. It is read as “does not belong to”.

Ex: Let $A = \{5, 10, 15, 20, \dots\}$, set of all multiples of 5.

Here $30 \in A$ but $24 \notin A$

Cardinality of a Set

The number of elements present in a set is called the **cardinal number, cardinality, or order** of a set. If A is a set, its cardinality is denoted by $n(A)$.

Ex : Let $A = \{5, 7, 9, 11, 13\}$. Then $n(A) = 5$.

Note : The cardinality of empty set is **zero** as it has no elements.

Representation of Sets

Sets can be represented in two ways:

- Roster Form or Tabular form
- Set Builder Form or rule method

Representation of Sets

Roster Form or Tabular form :

In this form, all the elements of the set are listed, separated by commas and enclosed between curly braces { }.

Ex :

$A = \{2, 4, 8, 16, \dots\}$, set of all positive powers of 2.

$B = \{a, e, i, o, u\}$, set of vowels

Representation of Sets

Set builder form or rule method :

Instead of listing the elements as in roster form, here we describe a set using a rule or condition. We represent the elements of a set X in this form by

$$X = \{x : (\text{condition followed by } x \text{ or properties of } x)\}$$

Ex :

$$A = \{x : x \text{ is a prime number between 2 and 10}\}$$

$$B = \{x : x^2 = 9, x \in \mathbb{Z}\}$$

Representation of Sets

Open , Closed and Semi open, Semi closed intervals :

$$A = \{x : 2 < x < 3, x \in \mathbb{R}\} = (2, 3)$$

$$B = \{x : 0 \leq x \leq 1, x \in \mathbb{R}\} = [0, 1]$$

$$C = \{x : -5 < x \leq 10, x \in \mathbb{R}\} = (-5, 10]$$

$$D = \{x : 5 \leq x < 10, x \in \mathbb{R}\} = [5, 10)$$

Types of sets

- **Empty set (Null set)** : A set with no elements is an empty set. It is denoted by $\{\}$ or ϕ . Ex : A = Set of all natural numbers less than 1
- **Singleton set (Unit set)** : A set with one element is a singleton set.
Ex : B = Set of even prime numbers , C = $\{0\}$
- **Finite set** : A set having finite or countable number of elements is called finite set. Ex : A = $\{10, 20, 30\}$

Types of sets

- **Infinite set** : A set containing an infinite number of elements is called an infinite set. Ex : $B = \text{Set of all integers divisible by 3}$
- **Equal sets** : Two sets are equal if they contain exactly the same elements.
Ex : $A = \{1, 2, 3\}$, $B = \{3, 1, 2\}$. Here A and B are equal sets.
- **Equivalent sets** : Sets which have same number of elements, but not necessarily the same elements.
Ex : $A = \{1, 2, 3\}$, $B = \{a, b, c\}$. Here A and B are equivalent sets. i.e. $A \sim B$

Types of sets

- **Subset :** Set A is said to be subset of B if every element of A is also an element of B . Ex : $A = \{1, 2, 3\}$, $B = N$. Then $A \subseteq B$.
- **Super set :** If A is a subset of B , then B is called the Super set of A .
i.e. If $A \subseteq B$, then $B \supseteq A$.
- **Proper Subset :** If A is a proper subset of B , then all elements of A are in B but B contains at least one element that is not in A .
Ex : $A = \{5, 7\}$, $B = \{5, 7, 8, 9\}$. Then $A \subset B$ and $A \neq B$.

Types of sets

- **Universal set** : A universal set is a set which contains all the elements of the other sets. It is denoted by U .

Ex : Let $A = \{a, b\}$, $B = \{c, d\}$, $C = \{e, f\}$. Then $U = \{a, b, c, d, e, f\}$

Here A , B and C are all subsets of U .

- **Power set** : A power set is defined as the set of all possible subsets of a given set, including the empty set and the set itself. Power set of the set A is denoted by $P(A)$. If A has 'n' elements, then $P(A)$ will have 2^n elements.

Ex : Let $A = \{2, 4, 6\}$. Then $P(A) = \{\emptyset, A, \{2\}, \{4\}, \{6\}, \{2, 4\}, \{4, 6\}, \{2, 6\}\}$.

Here A has 3 elements. So, $P(A)$ has $2^3 = 8$ elements.